# GUJARAT TECHNOLOGICAL UNIVERSITY

**Program Name: Diploma Engineering**

**Level: Diploma**

**Branch: Chemical Engineering**

**Subject Code: DI04005071**

**Subject Name: Petroleum Refining and Petroleum Technology**

| | |
|---|---|
| **w. e. f. Academic Year:** | 2025-26 |
| **Semester:** | 4th |
| **Category of the Course:** | Professional Elective - II |

| | |
|---|---|
| **Prerequisite:** | Student should have knowledge regarding Unit Operations and basics of Chemical Process Technology |
| **Rationale:** | India ranks among the top five refining nations globally and Petrochemicals, Derived primarily from hydrocarbons such as natural gas and crude oil, these chemical compounds are the building blocks of countless products and industries. A diploma chemical engineer has to apply relevant concepts to operations, equipment, safety protocols and environment considerations in refineries and petrochemical industries. A diploma chemical engineer need to be equipped with necessary skills and knowledge to work in petroleum refining and petrochemical plants, quality testing laboratories and related fields. This course is designed to develop such competency and skills. |

**Course Outcome:**

After Completion of the Course, Student will able to:

| No | Course Outcomes | RBT Level |
|---|---|---|
| 01 | Identify properties of crude petroleum and petroleum products | U, A |
| 02 | Explain fractionation of crude petroleum and treatment techniques | R, U |
| 03 | Apply refinery processes to maximize desired petro products | R, U, A |
| 04 | Explain manufacturing processes of petrochemicals | R, U |

**Revised Bloom's Taxonomy (RBT)*

**Teaching and Examination Scheme:**

| Teaching Scheme (in Hours) | | | Total Credits L+T+ (PR/2) | Assessment Pattern and Marks | | | | Total Marks |
|---|---|---|---|---|---|---|---|---|
| | | | | Theory | | Tutorial / Practical | | |
| L | T | PR | C | ESE (E) | PA(M) | PA(I) | ESE (V) | |
| 3 | 0 | 2 | 4 | 70 | 30 | 20 | 30 | 150 |

**Course Content:**

| Unit No. | Content | No. of Hours | % of Weightage |
|---|---|---|---|
| **Unit – I Basics of Petroleum and Refinery** | 1.1 Occurrence of petroleum<br>1.2 Composition of Petroleum<br>1.3 Classification of Petroleum<br>1.4 Types of Refineries<br>1.5 Refinery processes<br>1.5.1 Physical Changes<br>1.5.2 Chemical Changes<br>1.6 Refinery Products | 04 | 11 |
| **Unit– II Fractionation of Petroleum** | 2.1 Primary treatment of crude<br>(a) Dehydration and Desalting of crude oil<br>(b) Pipe still heaters<br>2.2 Atmospheric distillation of crude<br>2.3 Vacuum distillation of crude residue<br>2.4 Physical properties of petroleum products and its measurements:<br>(a) Petrol<br>(b) Diesel<br>(c) Kerosene<br>(d) Lubricant oil<br>(e) CNG and LPG<br>(f) Grease | 9 | 16 |
| **Unit– III Refinery Processes** | 3.1 Cracking<br>(a) Purpose of cracking<br>(b) Effect of temperature and pressure on Cracking<br>(c) Cracking methods<br>i. Thermal Cracking: Visbreaking, Delayed Coking and Fluid Coking<br>ii. Catalytic Cracking<br>iii. Fluidized Bed Catalytic Cracking<br>iv. Hydrocracking<br>(d) Compare thermal and catalytic cracking<br>3.2 Reforming<br>(a) Purpose of reforming<br>(b) Platforming<br>3.3 Hydrotreating | 10 | 20 |
| **Unit– IV Treatment** | 4.1 Purposes and methods of sulfur removal<br>4.2 Methods of sulfur removal | 11 | 24 |

| | | | |
|---|---|---|---|
| **Techniques** | (a) Doctor's sweetening<br>(b) Catalytic desulfurization<br>(c) MEROX treatment<br>4.3 Treatment of Kerosene by liquid $SO_2$ extraction<br>4.4 Solvent extraction processes<br>(a) Furfural extraction method (b) Phenol extraction method<br>4.5 Purpose of dewaxing<br>4.6 Dewaxing Techniques<br>(a) Dewaxing without solvent<br>(b) Dewaxing with solvent<br>i. Ketone dewaxing | | |
| **Unit – V Petrochemicals** | 5.1 Basics of petrochemicals: Definition, Classification and feedstocks for petrochemicals<br>5.2 Manufacturing of important C1 compounds<br>(a) Methanol<br>(b) Formaldehyde<br>5.3 Manufacturing of important C2 compounds<br>(a) Vinyl chloride<br>(b) Ethylene oxide<br>5.4 Manufacturing of important C3 compounds<br>(a) Propylene oxide<br>5.5 Chemicals from aromatics<br>(a) Manufacture of LAB | 11 | 29 |
| | **Total** | **45** | **100** |

**Suggested Specification Table with Marks (Theory):**

| Distribution of Theory Marks (in %) | | | | | |
|---|---|---|---|---|---|
| **R Level** | **U Level** | **A Level** | **N Level** | **E Level** | **C Level** |
| 27 | 24 | 19 | --------- | _____ | _____ |

*Where R: Remember; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create (as per Revised Bloom's Taxonomy)*

**References/Suggested Learning Resources:**

**(a) Books:**

| S.No. | Title of Book | Author | Publication with place, year and ISBN |
|---|---|---|---|
| 1 | Modern Petroleum refining Processes | B. K. Bhaskar Rao | Oxford and IBH, 2007 |

| | | | |
|---|---|---|---|
| 2 | Outlines of chemical Technology | M. Gopala Rao, Marshall Sitting | 3rd Edition East-West Press Pvt. Ltd, Delhi |
| 3 | Shreve's Chemical Process Industries | Austin G.T. | McGraw Hill publication – New Delhi, 5th edition |
| 4 | A Text on Petrochemicals | B. K. Bhaskar Rao | 2nd Edition, Khanna Publishers, Delhi, 1998 |
| 5 | Petroleum Refinery Engineering | W. L. Nelson | McGraw Hill, New York, 1958 |

**(b) Open-source software and website:**

1. https://onlinecourses.nptel.ac.in/noc23_ch64
2. https://www.e-education.psu.edu/fsc432/content/overview-refinery-products-and-processes
3. http://www.setlab.com/resources/refining/solvent-extraction-dewaxing/#1498503459530-b4e0a336-25dd
4. https://thepetrosolutions.com/solvent-extraction-process-in-petroleum-oil-refinery/
5. https://www.e-education.psu.edu/fsc432/content/dewaxing
6. https://thepetrosolutions.com/thermal-cracking-process-in-oil-refinery/
7. MIT Open course lecture available on Internet etc…
8. National Digital Library of India

**Suggested Practical List:**

| Sr. No. | Practical/Exercise (Course Outcomes in Psychomotor Domain according to NBA Terminology) | Approx. Hrs Required |
|---|---|---|
| 1 | Prepare a detail chart of modern refinery | 2 |
| 2 | Prepare a detail chart of petrochemical products | 2 |
| 3 | Determine flash point of petroleum product by Penskey Martin method. | 2 |
| 4 | Determine fire point of petroleum product by Penskey Martin method. | 2 |
| 5 | Measure softening point of petroleum product. | 2 |
| 6 | Measure Aniline point of petroleum product. | 2 |
| 7 | Determine penetration number of Grease. | 2 |
| 8 | Determine Carbon residue by Ram's bottom method. | 2 |
| 9 | Determine Carbon residue by conradson method | 2 |
| 10 | Measure smoke point of kerosene. | 2 |
| 11 | Measure cloud point of given petroleum product. | 2 |

| | | |
|---|---|---|
| 12 | Measures pour point of given petroleum product. | 2 |
| 13 | Measure initial & final boiling point of petroleum product. | 2 |
| 14 | Measure Viscosity of lube oil by Redwood /Saybolt/Engler | 2 |

**Suggested Activities for Students:**

1. Assignments
2. Technical Quiz/MCQ Test
3. Course/topic-based presentation
4. Market survey of various petrochemical products of different manufacturers and their comparison based on their specification, composition and cost
5. Preparation of non-working models/demo models of various ADU, VDU, Cracking units etc.
6. Preparation of power-point slides, which include videos, animations of various refining operations and processes

* * * * * * *